European Option Pricing

A Comparative Study of Binomial Trees and the Black–Scholes Model, with Real-Market Data and Machine-Learning Benchmarks

MAC F244 · Project proposal 31 October 2026

Problem statement. European options can be priced by a discrete-time no-arbitrage argument (the binomial tree) or by continuous-time stochastic calculus (the Black–Scholes model). The tree is known to converge to Black–Scholes as the number of steps grows, but the speed and regularity of that convergence, the flexibility of each model, and how well either describes real market prices are open practical questions. This project studies them through theory, controlled numerical experiments, real NSE option data, and machine-learning benchmarks. Our central question is:

> *For European options on Indian market underlyings, how do the binomial and Black–Scholes models compare in accuracy, speed, flexibility and real-market fit, and does the choice of* model *or the choice of* inputs *(especially volatility) matter more?*

Group members

Student	ID number
Adwait Eklavya	2024ADPS0825G
Mohammad Zaid Agha	2024ADPS0840G
Asvin Sunderiyal	2024ADPS0867G
Shivank Mishra	2024ADPS0823G
Pradyumn Venkata Rayasam	2023B5AD0973G

1 Project Aim and Scope

The aim is a reproducible computational study of European call and put pricing that connects the one-period replication argument, the multi-period Cox–Ross–Rubinstein (CRR) tree, risk-neutral geometric Brownian motion (GBM) and the Black–Scholes (BS) formula. Beyond reproducing prices, we will (i) quantify how and why the tree converges to BS, (ii) identify settings where the tree is more flexible than BS, (iii) test both models against market option prices for several assets and periods, and (iv) benchmark each part against a machine-learning (ML) model.

The baseline contract is $S_0 = 100$, $K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ year, zero dividend yield, with BS reference values $C_0 = 10.4506$ and $P_0 = 5.5735$. All methods are validated on this case before parameter sweeps. The scope is European vanilla options; American exercise, path-dependent payoffs, Monte Carlo pricing and transaction costs are outside the project.

2 Mathematical Models

2.1 Binomial model and risk-neutral valuation

Divide $[0, T]$ into N steps of length $\Delta t = T/N$. In the CRR model

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}}, \qquad q = \frac{e^{(r-\delta)\Delta t} - d}{u - d},$$

where δ is a continuous dividend yield and q the risk-neutral up-probability; the no-arbitrage condition $d < e^{(r-\delta)\Delta t} < u$ gives $0 < q < 1$. With terminal payoffs $C_T = (S_T - K)^+$ and $P_T = (K - S_T)^+$, backward induction gives

$$V_{n,j} = e^{-r\Delta t}\big[q\, V_{n+1,j+1} + (1-q)\, V_{n+1,j}\big].$$

This will be derived from a one-period replicating portfolio. We will also implement the Jarrow–Rudd (equal-probability) and Leisen–Reimer (Peizer–Pratt inversion) parametrisations, and compute tree Greeks (Δ, Γ, Θ) from the first nodes.

2.2 GBM and the Black--Scholes benchmark

Under the risk-neutral measure $\mathbb{Q}$,

$$dS_t = (r-\delta)S_t\, dt + \sigma S_t\, dW_t^{\mathbb{Q}}, \qquad S_T = S_0 \exp\Big[\big(r - \delta - \tfrac{1}{2}\sigma^2\big)T + \sigma\sqrt{T}\, Z\Big], \quad Z \sim \mathcal{N}(0,1).$$

Itô's lemma and a delta-hedged portfolio give the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r-\delta)S\frac{\partial V}{\partial S} - rV = 0,$$

whose solutions for a European call and put are

$$C_0 = S_0 e^{-\delta T}\Phi(d_1) - K e^{-rT}\Phi(d_2), \qquad P_0 = K e^{-rT}\Phi(-d_2) - S_0 e^{-\delta T}\Phi(-d_1),$$

$$d_1 = \frac{\ln(S_0/K) + (r - \delta + \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

Put–call parity, $C_0 - P_0 = S_0 e^{-\delta T} - K e^{-rT}$, is used as a validation identity. Closed-form Greeks (Δ, Γ, Vega, Θ, ρ) serve as references for the tree and ML estimates.

2.3 Convergence of the tree to Black--Scholes

In the tree, $\ln S_T$ is a sum of N i.i.d. two-point steps whose mean and variance match those of risk-neutral GBM to first order in Δt; by the central limit theorem the terminal distribution converges to the lognormal one, and the option price converges to the BS value. For CRR the

error is known to be $O(1/N)$ with an oscillating component that depends on where the strike falls between terminal nodes, producing different behaviour for even and odd N. We measure

$$e_N = \left|V_N^{\text{tree}} - V^{\text{BS}}\right|, \qquad e_N^{\text{rel}} = e_N / |V^{\text{BS}}|,$$

estimate the rate from a log–log fit, and test corrections: averaging N and $N{+}1$, BS-smoothing of the last step, and the Leisen–Reimer tree.

2.4 Volatility, implied volatility and dividends

Historical volatility is estimated from log-returns $R_t = \ln(S_t/S_{t-1})$ using rolling windows (30, 60, 252 days) and an exponentially weighted (EWMA) estimator. Implied volatility σ_{imp} is found by solving $V^{\text{BS}}(\sigma) = V^{\text{mkt}}$ with Newton's method (bisection as fallback), giving the volatility smile across strikes. For dividend-paying stocks, the tree applies discrete dividends on their actual dates, while BS uses the escrowed-dividend approximation $S_0 \mapsto S_0 - \sum_i D_i e^{-rt_i}$; the tree also allows time-varying σ and r per step.

3 Proposed Computational Methodology

All code will be written in Python (NumPy, SciPy, pandas, scikit-learn, Matplotlib), with each function documented against the equation it implements.

1. **Core engines and tests.** BS pricer with Greeks; vectorised CRR, Jarrow–Rudd and Leisen–Reimer trees. Unit tests: hand-computed one- and two-step trees, put–call parity, arbitrage bounds and limiting cases ($\sigma \to 0$, $T \to 0$).
2. **Convergence experiments.** Prices for every N from 1 to 1000; even/odd subsequences; log–log error rate; dependence of the error on moneyness, σ and T; effect of each correction; runtime against accuracy.
3. **Sensitivity analysis.** Sweeps of S, K, T, r and $\sigma \in [0.10, 0.60]$; price surfaces; tree Greeks against analytical Greeks, with explanation of discrepancies near the strike and maturity.
4. **Real market data.** NIFTY 50 index options and options on one high- and one low-volatility NSE stock (e.g. Reliance Industries, Hindustan Unilever), for several expiries and for a calm and a volatile period. Spot history from NSE / Yahoo Finance; risk-free rate from the RBI 91-day T-bill yield. Model prices are compared with market prices by moneyness and maturity, and implied-volatility smiles are constructed.
5. **Model flexibility and model risk.** Discrete dividends (tree vs escrowed BS) on a dividend-paying stock; time-varying σ in the tree; size of the price error caused by misestimating σ by ± 2–5% compared with the tree's discretisation error.
6. **Machine-learning benchmarks.** One machine-learning model per workstream, always evaluated against the corresponding mathematical model, with time-based train/test splits for market data.

4 Progress So Far

The BS pricer and a vectorised CRR tree have been implemented and validated on the baseline contract. BS gives $C_0 = 10.4506$ and $P_0 = 5.5735$, and put–call parity holds to machine precision. Table 1 shows the first convergence results.

N	CRR call	$\lvert e_N \rvert$ (call)	CRR put	Runtime
10	10.2534	1.97×10^{-1}	5.3764	0.14 ms
50	10.4107	3.99×10^{-2}	5.5336	0.44 ms
100	10.4306	2.00×10^{-2}	5.5536	0.74 ms
101	10.4680	1.74×10^{-2}	5.5909	0.36 ms
500	10.4466	4.00×10^{-3}	5.5695	2.01 ms
1000	10.4486	2.00×10^{-3}	5.5715	3.97 ms

Table 1: CRR prices for the baseline contract against BS ($C_0 = 10.4506$, $P_0 = 5.5735$).

For this at-the-money contract, even N underprices and odd N overprices, and the error is close to $2/N$. These results confirm the implementation and set the baseline for the experiments above.

5 Expected Outcomes

- Tree prices converging to BS at rate $O(1/N)$ with even/odd oscillation, and a quantified improvement from the corrections, with Leisen–Reimer expected to be the most accurate.
- Characterisation of where convergence is slowest (deep out-of-the-money, short maturity) and of the agreement between tree and analytical Greeks.
- A volatility smile in NSE data, showing a systematic weakness of both models' constant-σ assumption, and evidence on whether volatility estimation matters more than the number of steps.
- Settings (discrete dividends, time-varying volatility) where the tree is more accurate than BS.
- For each ML benchmark, a clear statement of where it matches the mathematical model and where it fails.
- Commented, tested, reproducible code, and a 3–4 page report.

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